

1. Write the inequalities in interval notation.

a. $x \leq 2$ $(-\infty, 2]$

c. $x > 16$ $(16, \infty)$

e. All real numbers $(-\infty, \infty)$

b. $-2 < x \leq 6$ $(-2, 6]$

d. $0 \leq x < 7$ $[0, 7)$

f. $x \geq -5$ $[-5, \infty)$

2. The graph represents the function $f(x) = -2x^2 + 8x$. Identify each of the properties listed.

Domain: $(-\infty, \infty)$

Range: $(-\infty, 8]$

Zeros: $(0, 0)$ $(4, 0)$

Interval of Increase: $(-\infty, 2)$

Interval of Decrease: $(2, \infty)$

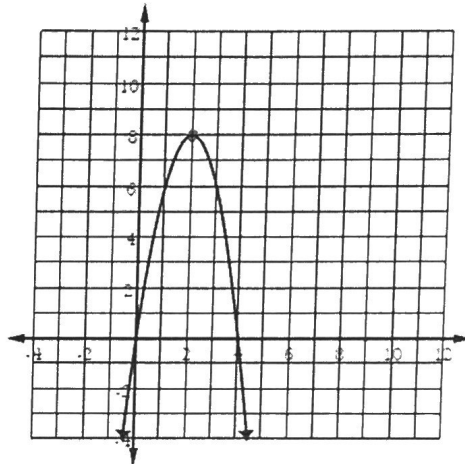
Vertex: $(2, 8)$

AOS: $x = 2$

x-intercepts: $(0, 0)$ $(4, 0)$

y-intercept: $(0, 0)$

Maximum or Minimum ?



3. The equation of a parabola is $f(x) = -2x^2 + 8x$. Determine the axis of symmetry and vertex algebraically. (Do not use your graphing calculator.) $V(2, 8)$ $x = 2$

4. Write the function $h(x) = 4x^2 + 12x + 8$ in factored form and determine the x-intercepts. (Use your graphing calculator.) x -int $(-1, 0)$ $(-2, 0)$ $h(x) = 4(x+1)(x+2)$

5. For each function, determine the characteristics indicated.

a. $f(x) = -3(x+2)(x-4)$

Direction parabola opens: down

x-intercepts: $(-2, 0)$ $(4, 0)$

y-intercept: $(0, 24)$

b. $f(x) = 5(x-3)^2 + 4$

Direction parabola opens: up

Vertex: $(3, 4)$

Axis of Symmetry: $x = 3$

y-intercept: $(0, 49)$

c. $f(x) = -3(x-5)(x+4)$

Direction parabola opens: down

x-intercepts: $(5, 0)$ $(-4, 0)$

y-intercept: $(0, 60)$

d. $f(x) = 6(x+4)^2 - 12$

Direction parabola opens: up

Vertex: $(-4, -12)$

Axis of Symmetry: $x = -4$

y-intercept: $(0, 84)$

6. Find the average rate of change on the specified interval

a. $[-2, 0]$ for $f(x) = -3(x+2)(x-4)$ $m = 12$

b. $[4, 6]$ for $f(x) = 5(x-3)^2 + 4$ $m = 20$

7. Write a function in vertex form that has these characteristics

- The function is quadratic and continuous, it opens upward, and has been translated down 3 units and right 3 units from the parent function $f(x) = x^2$. $y = (x-3)^2 - 3$
- The function is quadratic and continuous, it opens downward, has a dilation factor of $\frac{1}{2}$, and has been translated left 4 units. $y = -\frac{1}{2}(x+4)^2$
- The function is quadratic and continuous, it opens downward, has a dilation factor of 3, and has been translated up 1 unit. $y = -3x^2 + 1$
- The function is quadratic and continuous, it opens downward, has a dilation factor of 2, and has been translated right 1 unit and down 4 $y = -2(x-1)^2 - 4$

8. Graph each function with a minimum of 5 points. State the key characteristics.

a. $y = 3x^2 - 3$

Domain: $(-\infty, \infty)$

Range: $[-3, \infty)$

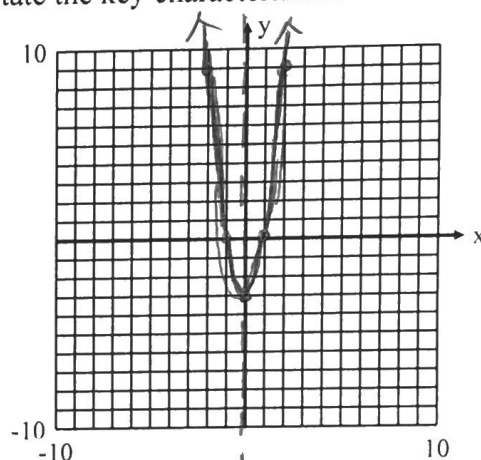
Vertex: $(0, -3)$

Axis of Symmetry: $x = 0$

x-intercepts: $(-1, 0)$ $(1, 0)$

y-intercept: $(0, -3)$

Concavity: up



b. $y = -3(x-1)^2 + 3$

Domain: $(-\infty, \infty)$

Range: $(-\infty, 3]$

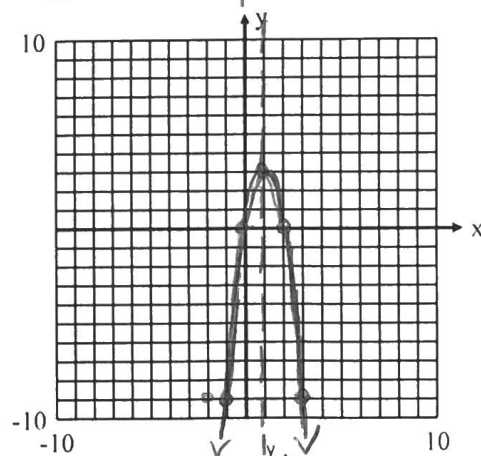
Vertex: $(1, 3)$

Axis of Symmetry: $x = 1$

x-intercepts: $(0, 0)$ $(2, 0)$

y-intercept: $(0, 0)$

Concavity: down



c. $y = (x-3)(x+5)$

Domain: $(-\infty, \infty)$

Range: $[-16, \infty)$

Vertex: $(-1, -16)$

Axis of Symmetry: $x = -1$

x-intercepts: $(3, 0)$ $(-5, 0)$

y-intercept: $(0, -15)$

Concavity: up

